

SupplyChain Atlas

Team Name: SupplyChain Atlas

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Team Size: 1

Problem Statement: Supply Chain Ontology and Governed Conversational Analytics

A Snowflake-native command center that connects operational signals to evidence-backed recovery decisions.

Problem Brief

Business problem

Carrier delays, sensor alerts, supplier reliability, and customer commitments sit in separate systems. Teams cannot quickly trace disruption to business exposure.

Target users

Control-tower managers, planners, supplier managers, and operations leaders who need a defensible recovery decision.

Solution

Atlas joins operational entities in governed Snowflake views, ranks risk, quantifies exposure, rehearses mitigation, and retains evidence.

Architecture Diagram

Operational sources

Suppliers, plants, parts, purchase orders, shipments, sensor events, and customer orders.

Snowflake processing

Core tables, Snowpark risk pipeline, SHIPMENT_RISK, and governed V_CONTROL_TOWER.

Decision layer

Streamlit Command Center, What-if Lab, Decision Receipt, Trust Center, and CoCo-assisted analysis.

Synthetic scenario: 120 shipments, 10 critical disruptions, 32 high-risk disruptions.

CoCo Workflow and Working Capabilities

Governed input

CoCo queries V_CONTROL_TOWER for the highest-risk shipment and cites the connected evidence.

Transparent processing

Snowpark combines delay, sensor condition, and supplier reliability into an explainable risk score.

Actionable output

The app prioritizes disruption, simulates recovered carrier days, and creates a Decision Receipt.

Recorded demo shows CoCo analysis, Command Center, What-if Lab, and Decision Receipt.

Impact Statement

Scenario impact

USD 541.76M revenue exposed, 10 critical shipments, 32 high-risk shipments, and 100% evidence coverage shown in the app.

Operational outcome

Operators rank recovery work by business impact and test a carrier intervention before committing resources.

Scalability

The same ontology can ingest live carrier, IoT, supplier, inventory, and customer-order feeds in Snowflake.

Delivery and Governance

Delivered assets

Public GitHub repository, SQL setup, governance views, synthetic generator, Snowpark pipeline, semantic model, and Streamlit app.

Trust controls

Synthetic data only, transparent risk factors, governed V_CONTROL_TOWER evidence, and a reviewable Decision Receipt.

Live deployment

SUPPLYCHAIN_ATLAS.CORE on SUPPLYCHAIN_HACK_WH with the live SupplyChain Atlas Streamlit app.

SupplyChain Atlas gives operators a traceable path from disruption signal to recovery decision.